

CSE 112 – Lab #3 Circle Class – Classes and Objects (page 843 - #8) with UML Diagram

Create a **circle class** in a separate header file that has the following private member variables:

1. Radius: a double
2. Pi: a double initialized to 3.14159

The class will have the following member functions:

1. Default Constructor that sets the radius to 0.0 (even though the program does not use it)
2. A Constructor that accepts the radius of the circle as an argument (the program will use this)
3. setRadius – a mutator function for the radius variable
4. getRadius – an accessor function for the radius variable
5. getArea – returns the area of the circle ($\text{area} = \pi * \text{radius} * \text{radius}$)
6. getDiameter – returns the diameter of the circle ($\text{diameter} = \text{radius} * 2$)
7. getCircumference – returns the circumference of the circle ($\text{circ} = 2 * \pi * \text{radius}$)

Requirements:

Using the main function on the next page, write a program that demonstrates the circle class by prompting the user for the circle's radius (with input validation), create a circle object using the overloaded constructor, and then report the area, diameter, and circumference with the output precision set to (2) two decimal places. Use a **separate header file for the class**. Add a UML Diagram for the class as a comment below main (10%).

The program will:

- Use a separate *.h file for the class and a *.cpp file for main.
- Use 3.14159 as the value for pi
- Include #ifndef and #endif directives as required
- Have a UML diagram of the Class with access specifiers, parameters, and return types (ref Figure 13-22 pg 826) as a comment below main.

```
// *****  
//  CLASS NAME  
// *****  
//  MEMBER VARIABLES  
//  
// *****  
//  MEMBER FUNCTIONS  
//  (with access specifiers, parameters,  
//  and return types)  
// *****
```

Grading will be based on meeting all of the lab requirements, adherence to programming standards, operation and accurate output, and programming style including white space and indentation.

Assignments submitted after the due date will incur a **grade penalty of three (3) points per day** after the first day, and **will not be accepted more than one (1) week late**.

Sample Main:

// Chapter 13, Programming Challenge 8: Circle Class

```
int main()
{
    double radius = 0; // To hold a radius

    // Get the radius.
    while (radius <= 0)
    {
        cout << "Enter the circle's radius: ";
        cin >> radius;
        if(radius <= 0)
            cout << "That is not a valid radius." << endl << endl;
    }

    Circle userCircle(radius);          // Create a Circle object with the specified radius.

    cout << setprecision(2) << fixed;

    // Display the circle's data.
    cout << "Radius: " << userCircle.getRadius() << endl;
    cout << "Area : " << userCircle.getArea() << endl;
    cout << "Diameter: " << userCircle.getDiameter() << endl;
    cout << "Circumference: " << userCircle.getCircumference() << endl;

    return 0;
}

// UML Diagram as a comment below main per Figure 13-24
// *****
//   CLASS NAME
// *****
//   MEMBER VARIABLES
//
// *****
//   MEMBER FUNCTIONS
//
// *****
```

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